

Assignment: Local Data Analysis of System Logs

1. Overview

This project automates the extraction and analysis of local system logs (Linux syslog format). It identifies system events, filters for critical errors, and provides visual insights into system health.

2. Features

- **Log Parsing:** Uses Regex to extract timestamps, hostnames, services, and messages.
- **Categorization:** Automatically classifies logs into **INFO** or **ERROR** based on keyword detection.
- **Visualization:** Generates pie charts for error distribution and bar charts for hourly activity spikes.
- **Reporting:** Exports a full processed dataset to HTML for easy browser viewing.

3. Setup & Automation

- **Dependencies:** Python 3.x, Pandas, Matplotlib, Seaborn.
- **Execution:** Run `python src/log_analyzer.py`.
- **Automation:** The script is designed to be scheduled via **cron** (Linux) for weekly log summaries.